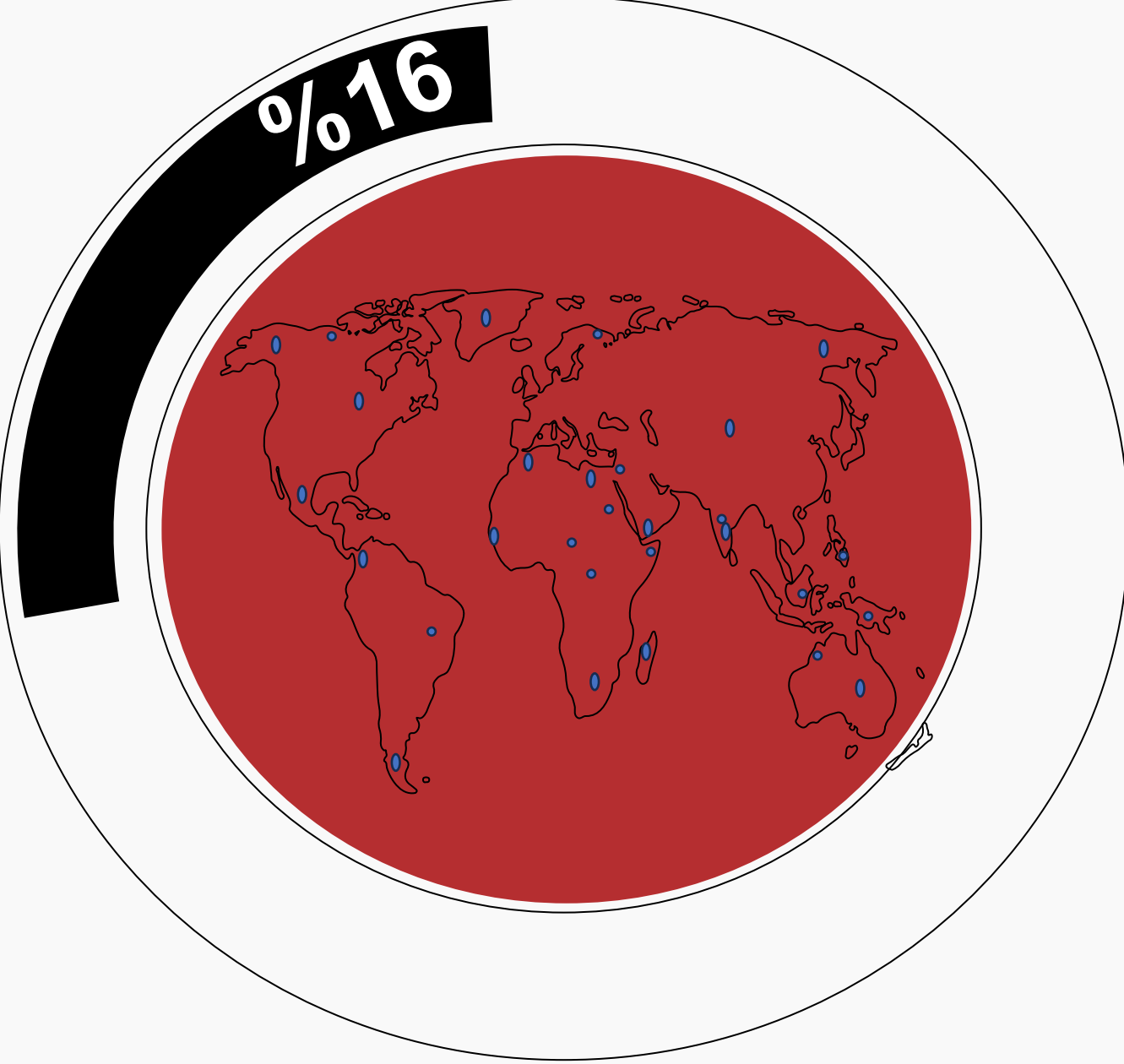


Bionic arm

Empowering
Possibilities

By the Bionic
Electronic Arm for
People with
Disabilities



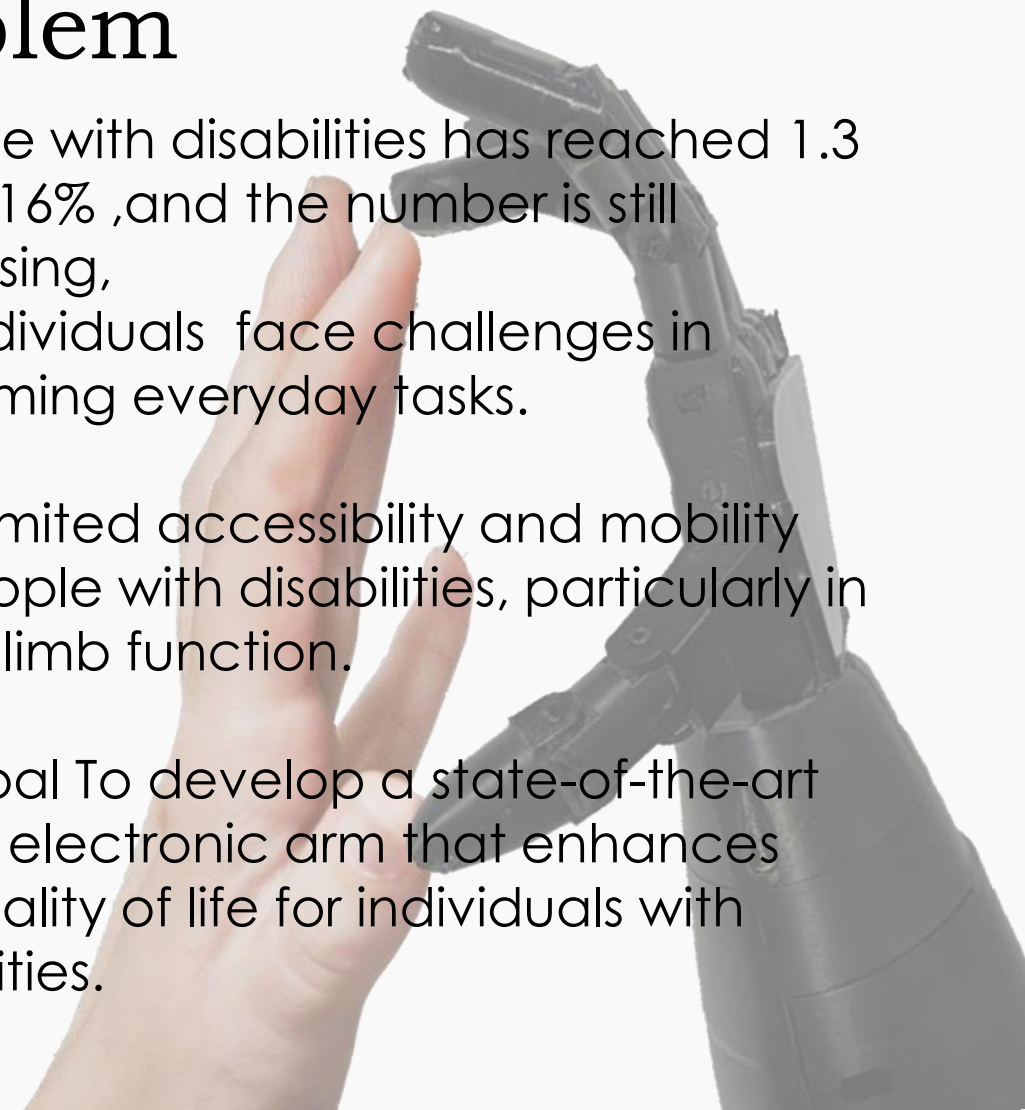


Problem

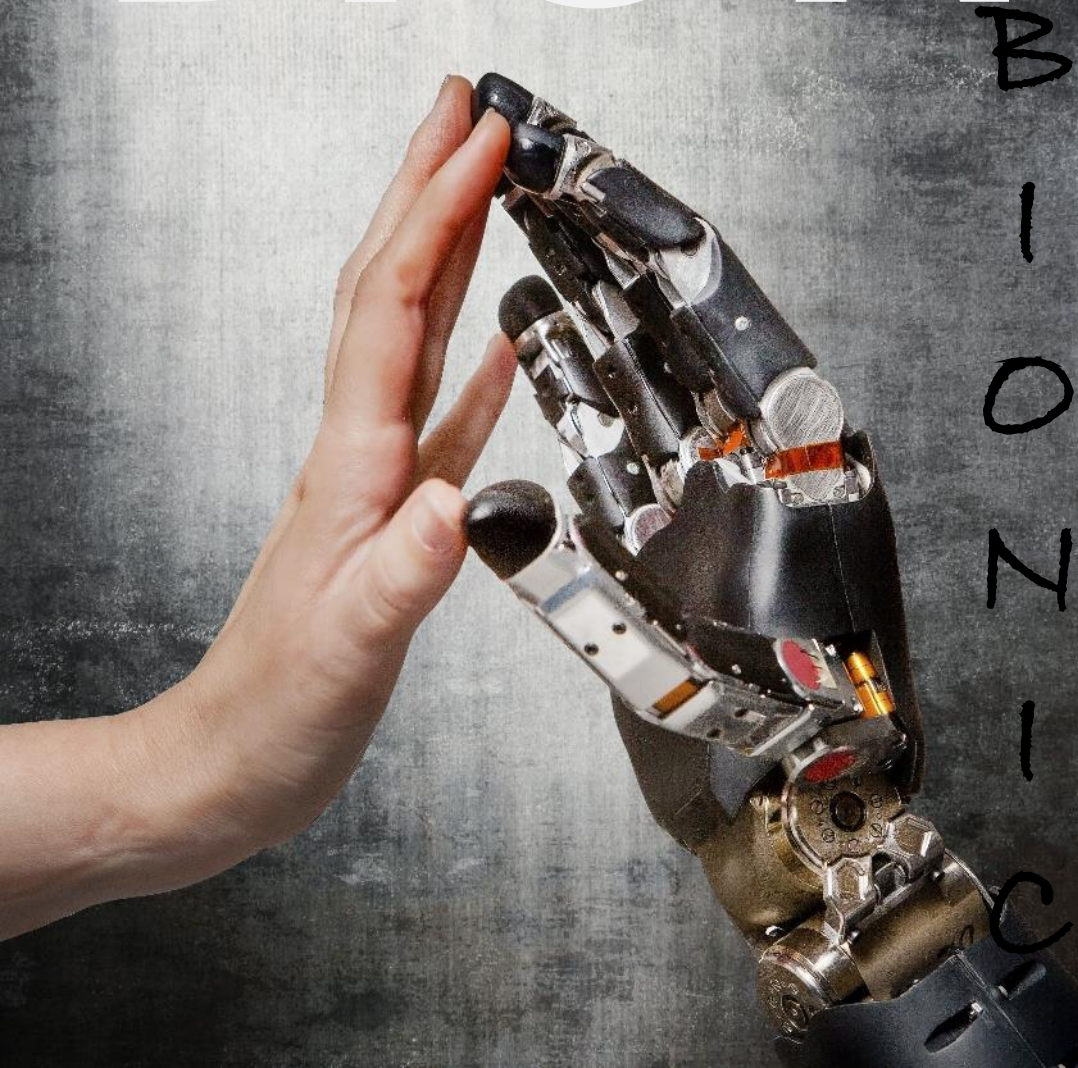
people with disabilities has reached 1.3 billion 16% ,and the number is still increasing,
The individuals face challenges in performing everyday tasks.

And Limited accessibility and mobility for people with disabilities, particularly in upper limb function.

Our goal To develop a state-of-the-art bionic electronic arm that enhances the quality of life for individuals with disabilities.



BIONIC ARM



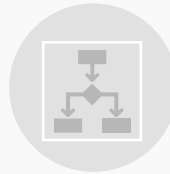
OUR SOLUTION



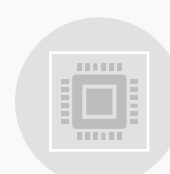
IOT App Handles the State Selection (switching modes) and Data Logging.



Technology is Advanced sensors, actuators, and neural interfaces.



3 phases ,3 prototype from 3d printer to carbon fiber functioning prototype the bionic electronic arm project.



Features: Customizable grip patterns, multi sensor fusion intuitive controls, lightweight design.

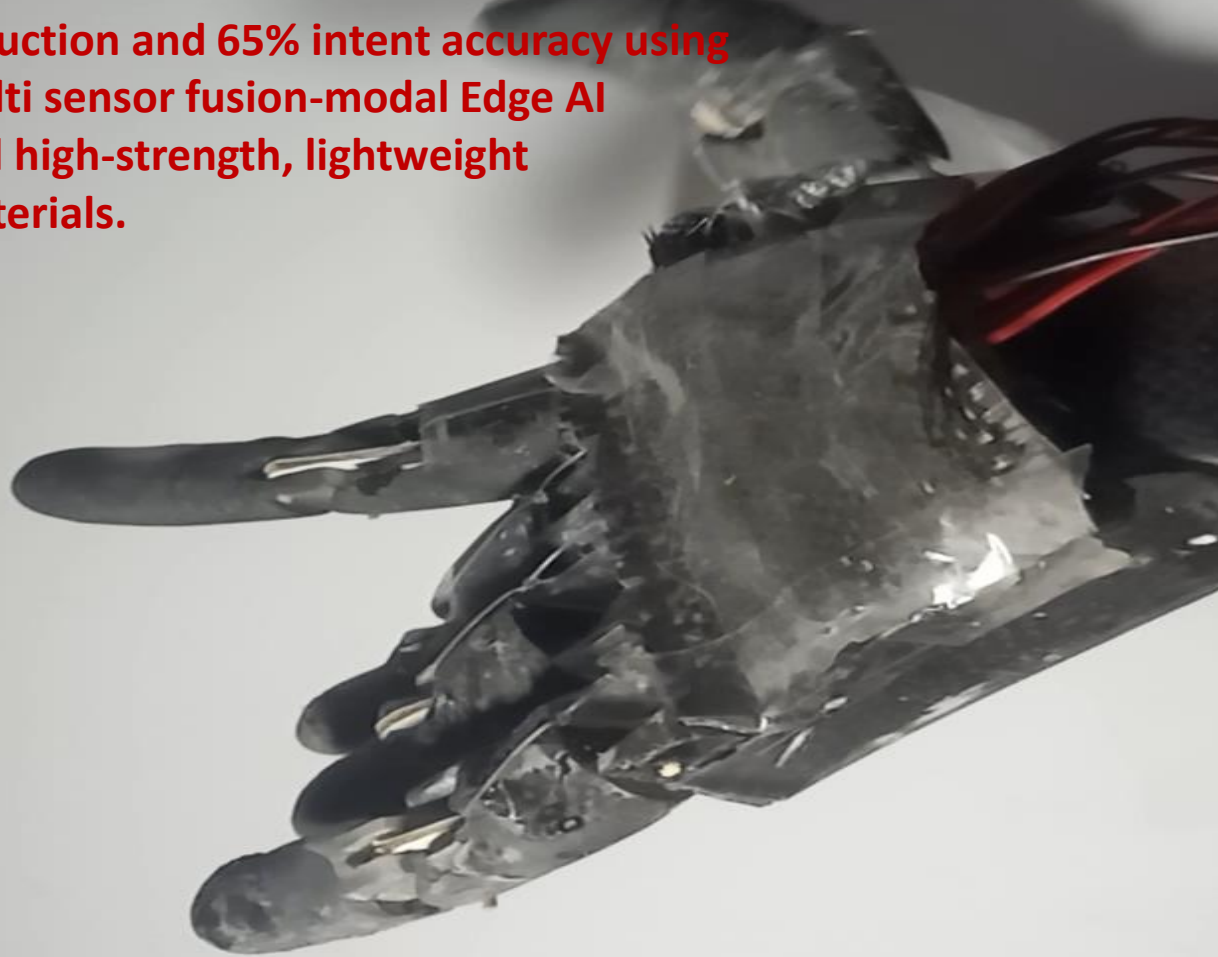


Benefits: Enhanced dexterity, improved functionality, increased independence.

4-Year R&D Journey: A longitudinal case study (2022–2026) involving 3 generational prototypes to eliminate control errors.

Iterative Evolution: Transitioned from basic (P1) to sensor-integrated models (P2) and finally to AI-driven carbon fiber design (P3).

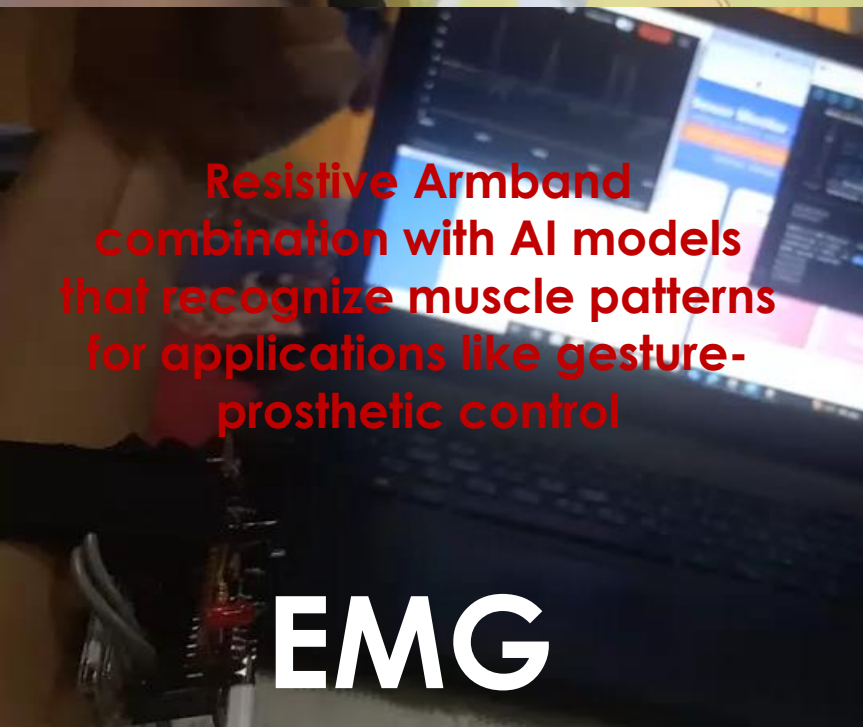
Optimized Results: Achieved 90% cost reduction and 65% intent accuracy using multi sensor fusion-modal Edge AI and high-strength, lightweight materials.





2x2 Matrix copper Resistive
Armband for gesture control

ARMBAND



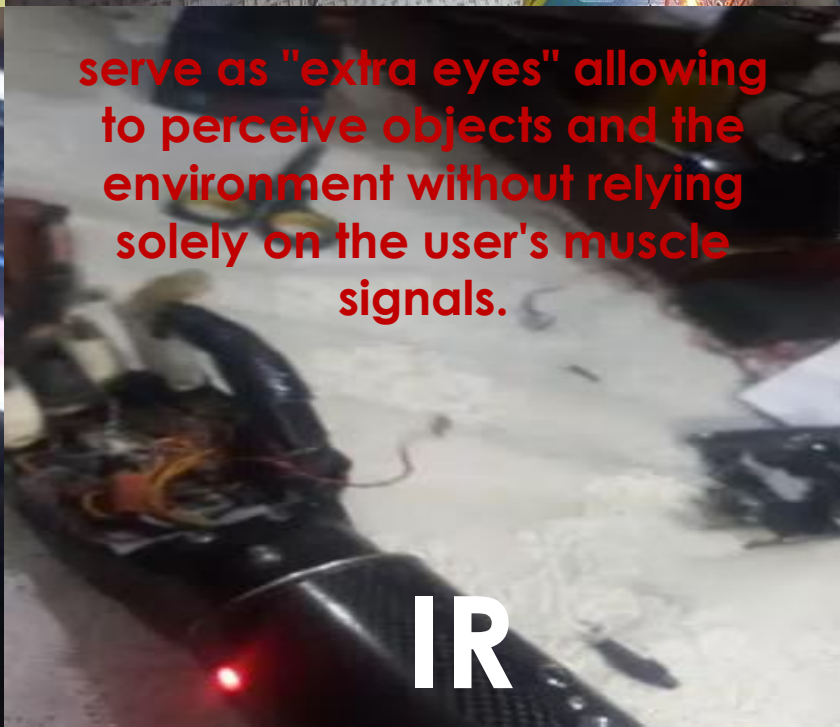
Resistive Armband
combination with AI models
that recognize muscle patterns
for applications like gesture-
prosthetic control

EMG



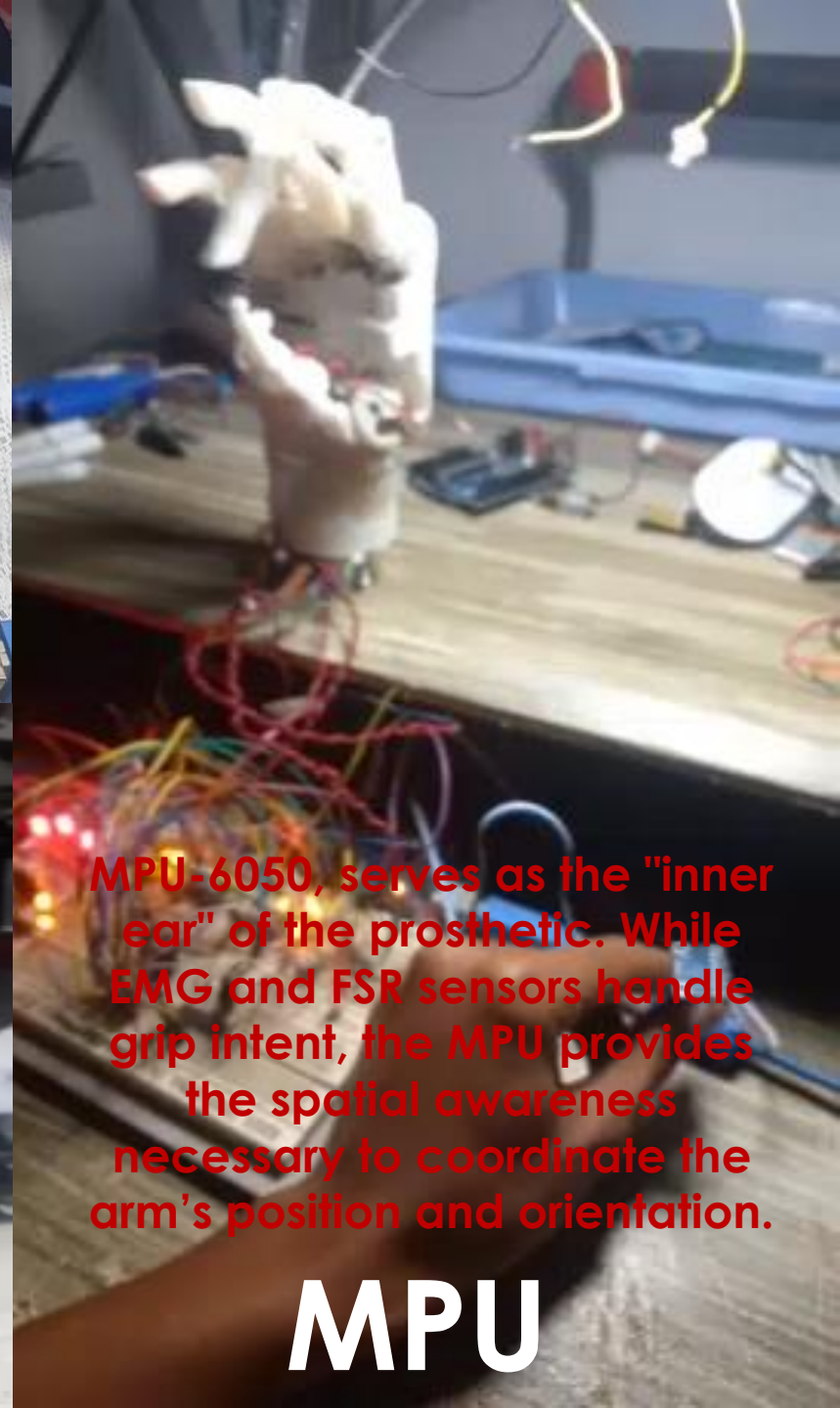
used for two primary purposes:
tactile feedback (sensing the
environment) and Force
Myography (sensing the user's
intent).

FSR



serve as "extra eyes" allowing
to perceive objects and the
environment without relying
solely on the user's muscle
signals.

IR



MPU-6050, serves as the "inner
ear" of the prosthetic. While
EMG and FSR sensors handle
grip intent, the MPU provides
the spatial awareness
necessary to coordinate the
arm's position and orientation.

MPU

12:34

VoLTE 26%



App



Bionic arm •

temperature

30

Palm control

wrist control



Pinkie control



Index control



Thumb control



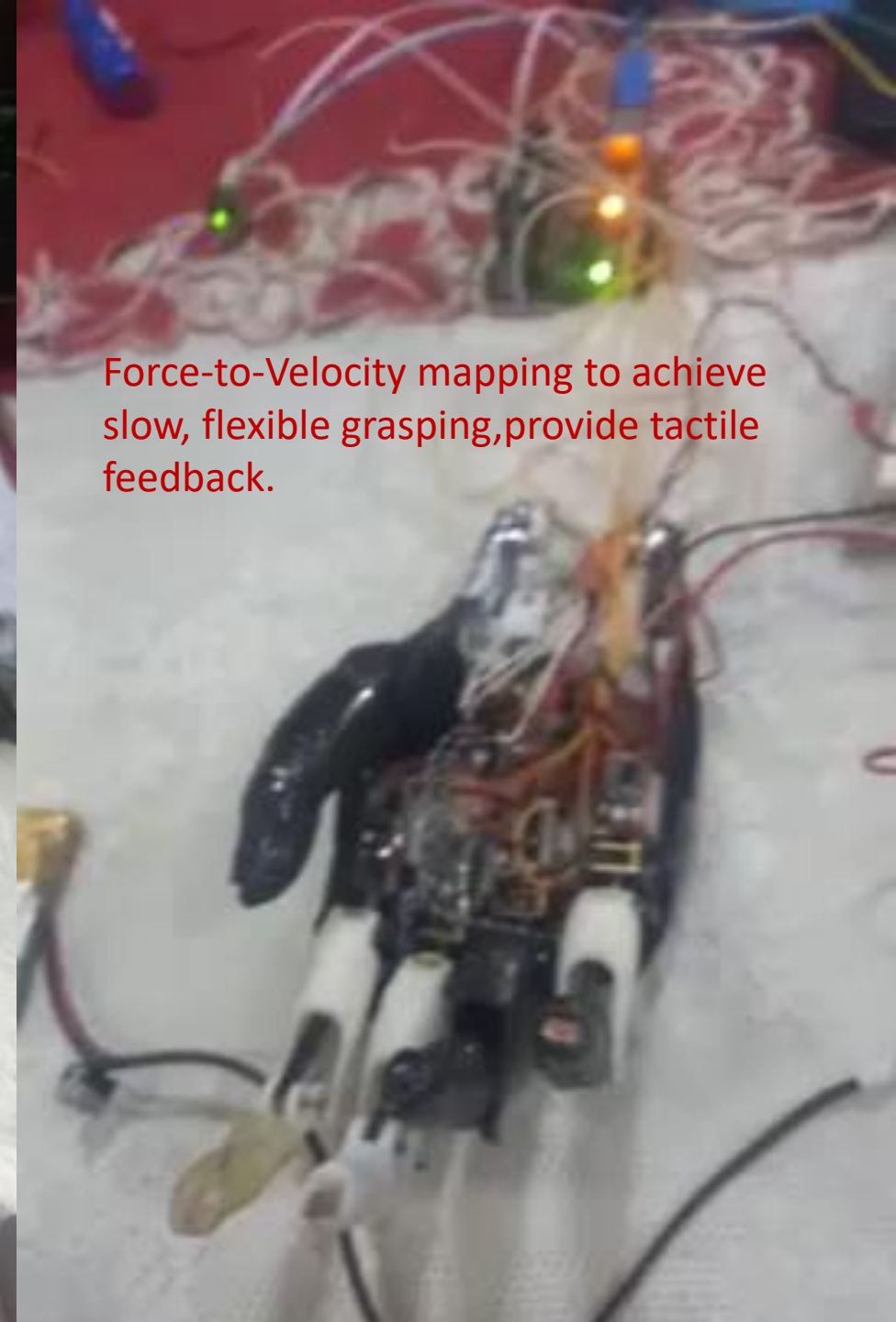
Middle ring control



app provides a user-friendly interface for users to control, easy adjustments, real-time signal visualization, and custom motion.



Force-to-Velocity mapping to achieve slow, flexible grasping, provide tactile feedback.



Lightweight and Comfortable

Weight about 350g

Made with carbon fiber and recycled materials for durability and sustainability.
Integrated PCB ensures efficient control without adding bulk.
Waterproof skin protects against moisture, allowing for versatile use

Accessible

Coast about 15000 EGP

to a wide range of users.
Value-driven design ensures affordability without compromising quality or functionality.
Designed to provide exceptional value for individuals seeking advanced assistive technology solutions.

Ambitious national development plan aimed at achieving sustainable economic growth and social development.

**Egypt vision
2030**

Opportunity for collaboration and partnership to contribute to Egypt's vision for a vibrant and inclusive society.

Integrating people with disabilities into society again

A hand is holding a black, textured, mechanical-looking object. The object has a rectangular base with a fine, grid-like texture and a more complex, angular top section. The hand is positioned behind the object, with fingers visible at the bottom and sides. The background is a light gray gradient.

**Thank you for
your attention!**